

LESSON 7-3 • SESSION 2 • PRACTICE • FORM A

Absolute Value as Total Distance

Learning target – I can use absolute value to find the total distance between a negative and a positive reading. • 6.NS.C.7c

REMEMBER

- **Opposite sides of 0:** add the absolute values. From -8 to 3 : $8 + 3 = 11$.
- **Same side of 0:** subtract the smaller absolute value from the larger. From -12 to -5 : $12 - 5 = 7$.
- A distance is **never negative**. Sketch a number line to see where 0 is.

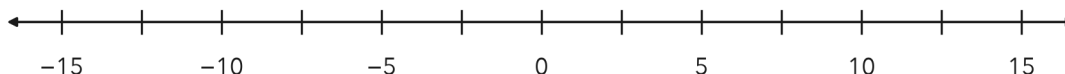
WORKED EXAMPLEHow far is it from -8 to 3 on a number line?**Step 1** 0 is between -8 and 3 , so they are on opposite sides.**Step 2** $|-8| = 8$ and $|3| = 3$.**Step 3** Add the distances: $8 + 3 = 11$.**Answer 11 units****WORDS** absolute value (*valor absoluto*) • distance (*distancia*) • opposite sides (*lados opuestos*)**PRACTICE**

show your work • label units

1. Find the distance between each pair of numbers.

a. -15 and 7 _____b. 3 and 19 _____c. -24 and -9 _____d. -6 and 4 _____

- 2.
- Reservoir Gauge.**
- A reservoir gauge read
- -12.5
- feet in August and
- 7.5
- feet in November. Zero is the normal water mark.

Hint Zero is between the readings, so add.

a. Plot both readings. Absolute values: August ____ November ____

b. Find the total swing. _____ ft

- 3.
- Garage Elevator.**
- An elevator goes from floor
- -4
- up to floor
- 11
- . How many floors does it travel?

Hint Count 4 floors up to 0, then 11 more.

Show your work.

Answer _____ floors

- 4.
- Drone Climb.**
- A drone climbs from
- 6
- m above the ground to
- 19
- m above the ground. How far did it climb?

Hint Both heights are above 0, so subtract.

Show your work.

Answer _____ m

- 5.
- Peak and Trench.**
- A peak rises
- 2.4
- km above the sea surface. A trench floor lies
- 3.6
- km below the sea surface. Find the vertical distance from the trench floor to the peak.

Answer _____ km

6. **Pump Move.** A well pump moved from -18 m to -41 m. Find the distance it moved. Why do you not add the absolute values?

Hint Both readings are below 0.

Show your work.

Answer _____ m

7. **Club Balance.** A club account moved from -67 dollars in May to 33 dollars in June. Find the total change.

Show your work.

Answer _____ dollars

8. **Mountain and Valley.** Trail 1 starts at -120 m in a valley and climbs to a peak at 340 m. Trail 2 runs from 90 m up to 340 m.

- a. Total climb of Trail 1: _____ m
 b. Total climb of Trail 2: _____ m
 c. Why is Trail 1 longer if both end at 340 m? _____

FIND THE ERROR

explain the mistake, then fix it

9. **Hugo's Work.** To find the swing from -9 degrees to 24 degrees, Hugo subtracted 9 from 24 and wrote 15 degrees. Explain his mistake. Then find the correct swing.

TEST PRACTICE

choose the best answer

10. The distance from -20 to -5 is –
A 25 **B** 15 **C** -15 **D** 5
11. When two readings are on opposite sides of 0, you –
A subtract their absolute values **B** add their absolute values
C multiply them **D** use only the larger one

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

100 N	59 Q	20 D	6 T	250 E	34 Z	13 S	15 I	23 A	460 C
#2b	#3	#4	#5	#6	#7	#8a	#8b		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____